

Chapter 2

Timescales of Thermal Control

Steady, Quasi-Steady, and Transient Regimes

Matching Metamaterial Capabilities to Process Timescales

Thermal Photonics: A Unified Framework for Engineering Heat Flux
Book 3 of the Open-Source Engineering Optics Trilogy

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Learning Objectives

After completing this chapter, the reader will be able to:

- LO-02.1** State the fundamental steady-state limitation of transformation thermotics: metamaterials control flux *direction* but not absolute temperature when boundary conditions are fixed.
- LO-02.2** Define thermal diffusivity α_{th} and derive the characteristic thermal time constant $\tau_{\text{th}} = L^2/\alpha_{\text{th}}$ for a system of length L .
- LO-02.3** Distinguish steady-state, transient, and quasi-steady-state regimes and identify the metamaterial capabilities available in each.
- LO-02.4** Apply the Fourier number $\text{Fo} = \alpha_{\text{th}} t/L^2$ and Biot number $\text{Bi} = hL/\kappa$ to classify a given semiconductor process into its temporal regime.
- LO-02.5** Evaluate the design criterion τ_{proc} vs. τ_{th} for cyclic semiconductor fabrication processes and determine whether passive metamaterial control or active feedback is required.
- LO-02.6** Locate a given application on the timescale landscape map (Figure 02.2) and use the map to select the appropriate analysis method.
- LO-02.7** Construct a dynamic thermal network incorporating thermal capacitance elements and extract system time constants via eigenvalue analysis.
- LO-02.8** Explain how Γ becomes time-dependent during rapid thermal processing and why the Mode-Path Matrix must be evaluated at multiple process snapshots.

Abbreviations and Nomenclature

This chapter introduces temporal-regime concepts building on the foundational vocabulary of Chapter 1.

Acronyms

Abbreviation	Meaning
ALD	Atomic Layer Deposition
BBR	Black-Body Radiation (shift, in optical clocks)
BC	Boundary Condition
Bi	Biot Number

Abbreviation	Meaning
BTE	Boltzmann Transport Equation
CVD	Chemical Vapor Deposition
DR	Design Rule (numbered as DR-02.n in this chapter)
DR-T	Design Rule — Topological series (Ch. 4)
ELA	Excimer Laser Annealing
EUV	Extreme Ultraviolet (lithography)
Fo	Fourier Number
FPA	Focal Plane Array
IC	Integrated Circuit
LSA	Laser Spike Annealing
LWIR	Long-Wave Infrared (8–14 μm)
MPM	Mode-Path Matrix (Ch. 4)
MWIR	Mid-Wave Infrared (3–5 μm)
PEB	Post-Exposure Bake
QKD	Quantum Key Distribution
RTP	Rapid Thermal Processing
SC	Superconducting
TSV	Through-Silicon Via
WE	Worked Example (numbered as WE-02.n)

Principal Symbols

Symbol	Meaning	SI Unit
α_{th}	Thermal diffusivity ($\kappa/\rho c_p$)	$\text{m}^2 \text{s}^{-1}$
Bi	Biot number (hL/κ)	—
c_p	Specific heat capacity	$\text{J kg}^{-1} \text{K}^{-1}$
C_{th}	Thermal capacitance ($\rho c_p V$)	J K^{-1}
δ_{pen}	Thermal penetration depth	m
Fo	Fourier number ($\alpha_{\text{th}} t/L^2$)	—
G	Thermal conductance	W K^{-1}
G_{cond}	Conduction conductance	W K^{-1}
G_{rad}	Radiation conductance	W K^{-1}
Γ	Mode dominance ratio ($G_{\text{rad}}/G_{\text{cond}}$)	—
h	Heat transfer coefficient	$\text{W m}^{-2} \text{K}^{-1}$
κ	Thermal conductivity	$\text{W m}^{-1} \text{K}^{-1}$
$\boldsymbol{\kappa}$	Thermal conductivity tensor	$\text{W m}^{-1} \text{K}^{-1}$
L	Characteristic length	m
L_{diff}	Thermal diffusion length ($\sqrt{\alpha_{\text{th}} \tau}$)	m
R_{th}	Thermal resistance	K W^{-1}
T	Temperature	K
τ_{th}	Thermal time constant ($R_{\text{th}} C_{\text{th}}$ or L^2/α_{th})	s
τ_{proc}	Process time constant	s
ρ	Mass density	kg m^{-3}
ε	Emissivity	—
σ_{SB}	Stefan–Boltzmann constant	$\text{W m}^{-2} \text{K}^{-4}$

Subscript and Superscript Conventions

Notation	Convention
cond, rad	Conduction and radiation modes
eff	Effective (combined or homogenized) value
ss	Steady state
tr	Transient
qs	Quasi-steady state
th	Thermal (as in τ_{th} , α_{th})
proc	Process (as in τ_{proc})
$\parallel$, $\perp$	Parallel and perpendicular to a reference axis

Design rules follow the DR-02.n convention. Worked examples use WE-02.n. Learning objectives use LO-02.n.

Chapter Roadmap and Deliverables

This chapter is organized around a single engineering deliverable:

A temporal classification framework—based on the Fourier number Fo , the Biot number Bi , and the timescale ratio $\tau_{\text{proc}}/\tau_{\text{th}}$ —that determines which metamaterial capabilities are available for a given process and how Γ evolves in time, expressed as a timescale landscape map spanning classical optics through quantum systems.

Figure 02.1 shows the chapter workflow spine.

Chapter 2 Roadmap

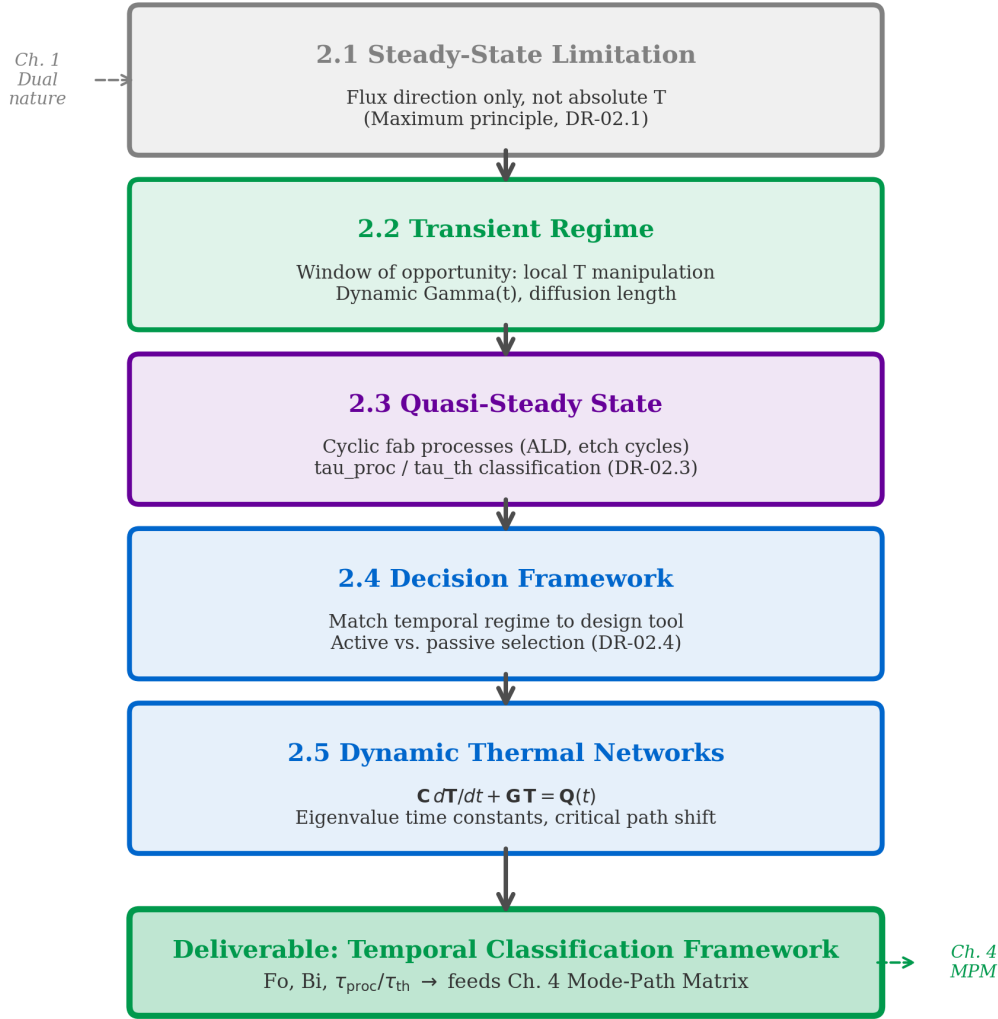


Figure 02.1: Chapter 2 roadmap. The application landscape (§02.1) motivates the three temporal regimes. The steady-state limitation (§02.2) constrains metamaterial capabilities; the transient regime (§02.3) opens a window for local temperature manipulation. Quasi-steady-state (§02.4) characterizes cyclic fab processes. The decision framework (§02.5) maps regimes to tools. Dynamic thermal networks (§02.6) extend the Mode-Path Matrix to time-dependent systems.

Pain Point

Thermal metamaterial literature overwhelmingly presents steady-state demonstrations: a thermal cloak that bends heat flux, a concentrator that focuses it. But real engineering systems—from LWIR lens modules warming up after power-on, through RTP wafers ramping to 1000°C in seconds, to superconducting qubit gate pulses lasting 20 ns—operate across *fourteen orders of magnitude* in process timescale. Engineers who design thermal management without identifying which temporal regime their process occupies risk deploying devices that work beautifully in steady-state simulation but fail in production because time matters.

Executive Summary. If you take away three quantities from this chapter: (1) $Fo = \alpha_{th} t / L^2$ classifies your temporal regime, (2) $Bi = hL / \kappa$ determines surface-vs-interior temperature uniformity, and (3) τ_{proc} / τ_{th} tells you which metamaterial tools are available and whether Γ must be tracked dynamically. Engineers who skip this classification step risk designing solutions for the wrong physics.

02.1 The Application Landscape: Where Does Your Process Live?

Before developing the mathematical framework, it is instructive to survey the landscape of thermal engineering across characteristic timescales. Table 02.4 catalogs representative applications from femtosecond laser processing ($\tau_{proc} \sim 100$ fs) to continuous furnace operation ($\tau_{proc} \rightarrow \infty$), organized by temporal regime. Table 02.5 traces the same framework from classical IR optical systems through semiconductor fabrication to quantum technology.

Table 02.4: Application landscape: characteristic timescales in thermal engineering, sorted by τ_{proc} from shortest to longest. The regime classification follows DR-02.3. Superscript notes define the τ_{th} reference geometry.

Application	τ_{proc}	τ_{th}	$\tau_{\text{proc}}/\tau_{\text{th}}$	Regime	Design Approach
Transient ($\tau_{\text{proc}}/\tau_{\text{th}} < 0.1$): Full dynamic analysis required					
Femtosecond laser anneal	100 fs	11 μs^a	10^{-8}	Trans.	Non-equilibrium; BTE
Excimer laser crystallization	20–50 ns	11 μs^a	10^{-3}	Trans.	Melt-front dynamics
SC qubit gate pulse	20–100 ns	1 μs^b	0.02–0.1	Trans.	Quasiparticle heat; Ch. 14
EUV resist (per pulse)	20 μs^c	6.8 ms ^a	3×10^{-3}	Trans.	Pulse-averaged model
Laser spike anneal	0.2–1 ms	6.8 ms ^a	0.03–0.15	Trans.	Transient shield (WE-02.3)
Microbolometer FPA frame	10 ms	2–5 ms ^d	2–5	Interm.	Pixel thermal response
Intermediate ($0.1 < \tau_{\text{proc}}/\tau_{\text{th}} < 10$): Full transient FEA; C_{th} critical					
Plasma etch RF cycling	2–20 ms	6.8 ms ^a	0.3–3	Interm.	Wafer C_{th} critical
Trapped-ion laser cooling cycle	1–100 μs	10 μs^e	0.1–10	Interm.	Ion trap thermal; Ch. 14
Wafer chuck zone heating	0.5–5 s	0.1–1 s ^f	1–10	Interm.	Active PID + passive
LWIR Ge lens power-on	5–30 s ^g	18 s ^h	0.3–1.7	Interm.	Focus drift; athermalize
MWIR missile seeker dome	2–10 s	5–15 s ⁱ	0.2–2	Interm.	Aero heating; dynamic OPD
Optical clock HVAC perturbation	10–60 min	25 min ^j	0.4–2.4	Interm.	Shield servo; Ch. 14
Satellite IR telescope (orbit)	90 min ^k	20–60 min ^l	1.5–5	Interm.	Orbital cycling; active + passive
Quasi-steady / Steady ($\tau_{\text{proc}}/\tau_{\text{th}} > 10$): Static analysis sufficient					
ALD cycle (wafer)	4.2 s	6.8 ms ^a	620	Quasi-ss	Steady-state (WE-02.2)
Post-exposure bake (PEB)	60–90 s	6.8 ms ^a	10^4	Steady	Bake uniformity
CVD deposition	60–600 s	6.8 ms ^a	10^4	Steady	Static Γ
Solder reflow	30–60 s	6.8 ms ^a	10^3	Quasi-ss	Ramp profile
Furnace oxidation	30–120 min	6.8 ms ^a	$> 10^5$	Steady	Equilibrium
LWIR lens module (steady operation)	continuous	18 s ^h	$\gg 10$	Steady	Athermalization; DR-01.3
3D IC package (steady)	continuous	0.1 s ^m	$\gg 10$	Steady	Network MPM; Ch. 4
Dilution fridge base stage	continuous	100 s ⁿ	$\gg 10$	Steady	Cryogenic MPM; Ch. 14
Optical clock BBR shield	continuous	25 min ^j	$\gg 10$	Steady	$\delta T < 15$ mK; Ch. 14

^aSi wafer through-thickness ($L = 775 \mu\text{m}$, $\alpha_{\text{th}} = 88 \text{ mm}^2/\text{s}$). ^bQubit chip local ($L \sim 30 \mu\text{m}$). ^c50 kHz pulse period. ^dVOx pixel ($L \sim 1 \mu\text{m}$, $\alpha_{\text{th}} \sim 0.5 \text{ mm}^2/\text{s}$). ^eTrap electrode-ion ($L \sim 100 \mu\text{m}$). ^fChuck zone thermal mass with PID. ^gEnvironmental slew time after power-on. ^hGe lens $\tau_{\text{th}} = R^2/\alpha_{\text{th}} = (25 \times 10^{-3})^2/(34 \times 10^{-6}) \approx 18$ s. ⁱDome wall ($L \sim 3$ mm, ZnS or sapphire). ^jCu shield $\tau \approx 25$ min (Ch. 14, WE-14.4). ^kLEO orbit period. ^lTelescope primary mirror thermal mass. ^mDie-substrate-heatsink chain. ⁿMXC thermal mass with 20 μW cooling.

Table 02.5: Classical-to-quantum progression: representative systems with characteristic timescales, mode dominance ratio Γ , and temporal regime. The framework applies identically across all domains.

Domain	System	τ_{proc}	Γ	Regime	Key Temporal Issue
<i>Classical IR Optics (Ch. 1, 10–11)</i>					
LWIR imaging	Ge lens, steady	Continuous	10^{-3}	Steady	Static thermal defocus (DR-01.3)
LWIR imaging	Ge lens, power-on	5–30 s	10^{-3}	Interm.	Focus drift during warm-up
MWIR seeker	Dome, aero heating	2–10 s	0.01–0.5	Interm.	Dynamic Γ ; transient OPD
Thermal imager	Microbolometer FPA	10 ms	~ 1	Interm.	Scene-induced transient per pixel
<i>Semiconductor Fabrication (Ch. 5–9, 14–15)</i>					
Front-end	Furnace oxidation	30–120 min	10^{-5} – 10^{-1}	Steady	Static Γ ; wall radiation
Front-end	RTP anneal	1–10 s	$10^{-5} \rightarrow 10^{-1}$	Interm.	Dynamic $\Gamma(t)$; mode transition
Front-end	Laser spike anneal	0.2–1 ms	$10^{-5} \rightarrow 0.6$	Trans.	Transient shield design
Litho	EUV (per pulse)	20 μs	10^{-6}	Trans.	Overlay thermal budget
Litho	Post-exposure bake	60–90 s	10^{-4}	Steady	Bake uniformity $\rightarrow$ CD
Back-end	3D IC steady	Continuous	10^{-5}	Steady	TSV bottleneck; Ch. 4 MPM
<i>Quantum Technology (Ch. 14)</i>					
SC qubits	Fridge base stage	Continuous	10^{-10}	Steady	Kapitza bottleneck
SC qubits	Gate pulse heating	20–100 ns	10^{-10}	Trans.	Quasiparticle generation
SC qubits	Cooldown (300 K)	12–48 h	$10^{-10} \rightarrow 1$	All 3	Γ traverses 10 decades
Trapped ions	Laser cooling	1–100 μs	0.1–10	Interm.	Mixed-mode; both modes matter
Opt. clock	BBR shield, steady	Continuous	~ 1	Steady	$\delta T < 15$ mK
Opt. clock	HVAC perturbation	10–60 min	~ 1	Interm.	Shield τ vs. HVAC cycle
QKD terminal	Detector interface	Continuous	0.5–5	Steady	Dark count rate thermal

02.1.1 Three Patterns from the Landscape

Three patterns emerge immediately from Tables 02.4 and 02.5:

Pattern 1: Silicon’s high diffusivity makes most fab steps quasi-steady through the wafer thickness. With $\alpha_{\text{th}} = 88 \text{ mm}^2/\text{s}$, a $775 \text{ }\mu\text{m}$ wafer has $\tau_{\text{th}} \approx 6.8 \text{ ms}$ —far shorter than nearly all process cycle times. The thermal design challenge in semiconductor fabrication is therefore not the wafer bulk but the *wafer-chuck* and *wafer-environment* interfaces, which have much longer time constants. This insight drives the interface-focused design philosophy of Chapters 4–9.

Pattern 2: Quantum systems exhibit the widest temporal range. A single superconducting quantum computer contains elements spanning from 20 ns gate pulses (transient) through continuous base-stage cooling (steady), with cooldown from 300 K traversing all three regimes over 12–48 hours as Γ spans ten decades. No other engineering domain requires such a broad temporal range—motivating the dedicated treatment in Chapter 14.

Pattern 3: Classical IR optics live in the intermediate regime most often. Germanium and chalcogenide lens elements have thermal time constants of 5–50 s (Table 02.4, note *h*), placing power-on transients and environmental perturbations squarely in the intermediate regime. This is precisely where neither steady-state athermalization nor simple pulse

analysis suffices—full transient modeling including the Fourier–Snell bridge (Chapter 1, §01.4) is required to predict focus drift and thermal aberration dynamics.

These observations motivate the mathematical framework developed in the remainder of this chapter.

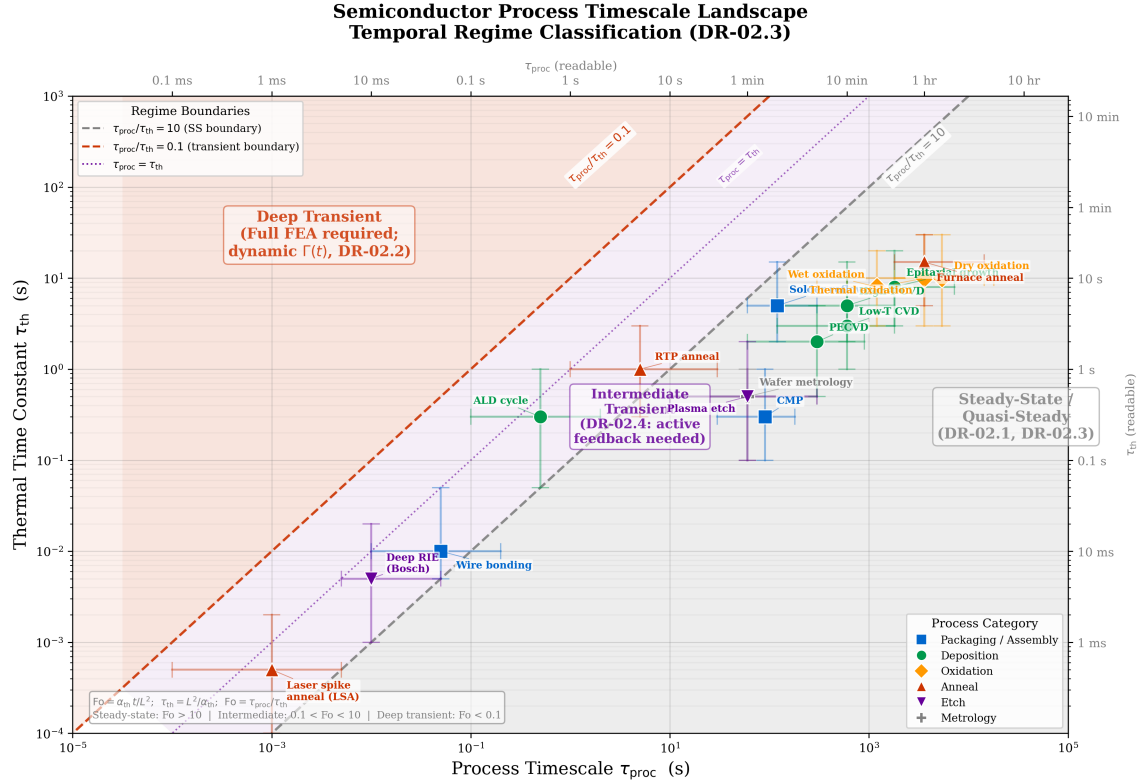


Figure 02.2: Timescale landscape map. Each application from Tables 02.4 and 02.5 is plotted on τ_{proc} vs. τ_{th} axes (both log scale). Diagonal lines at $\tau_{proc}/\tau_{th} = 0.1, 1$, and 10 separate the three temporal regimes. Markers are color-coded by domain: blue circles = classical IR optics, green squares = semiconductor fabrication, purple diamonds = quantum systems. This is the temporal analog of the Γ landscape (Ch. 1, Fig. 01.12; Ch. 14, Fig. 14.11).

Transient Regime

Practical Note: Choosing L for τ_{th} Estimation

The characteristic length L in $\tau_{th} = L^2/\alpha_{th}$ must match the *thermal transport path under analysis*, not the overall component size. Common choices:

- **Through-wafer conduction:** L = wafer thickness ($775 \mu\text{m}$ for 300 mm Si).
- **Radial heat flow in a lens:** L = lens radius (rim-to-center path).
- **Across a gap or TIM:** L = gap width or bond-line thickness.
- **Die-to-heat-sink (packaging):** L = total stack height (die + TIM + substrate + TIM2).

Misidentifying L is the single most common source of regime misclassification. When in doubt, evaluate τ_{th} for each distinct path segment separately and compare with τ_{proc} . The footnotes in Table 02.4 illustrate this practice for every listed application.

Conduction Mode (Diffusionics)

Temporal regimes in conduction: The heat diffusion equation $\rho c_p \partial T / \partial t = \nabla \cdot (\boldsymbol{\kappa} \cdot \nabla T) + \dot{q}$ is inherently time-dependent. Fourier’s law gives the instantaneous flux; the time derivative determines the regime. Transformation thermotics manipulates $\boldsymbol{\kappa}$ but does *not* alter ρc_p —with profound implications for transient performance.

Radiation Mode (Thermal Photonics)

Temporal regimes in radiation: Radiative transfer responds essentially instantaneously to surface temperature changes. The “transient” behavior of radiative systems originates from the thermal inertia of the emitting bodies, not from the radiation itself. The Stefan–Boltzmann T^4 nonlinearity makes $G_{\text{rad}} = 4\varepsilon\sigma_{\text{SB}}T^3A$ strongly temperature-dependent, so Γ varies during any temperature transient.

02.2 Steady State: What Metamaterials Can and Cannot Do**02.2.1 The Governing Equation at Steady State**

When all time derivatives vanish ($\partial T / \partial t = 0$), the heat equation reduces to:

$$\nabla \cdot (\boldsymbol{\kappa} \cdot \nabla T) + \dot{q} = 0 \quad (02.1)$$

Without internal heat generation ($\dot{q} = 0$), this is Laplace’s equation for the temperature field. Solutions are completely determined by boundary conditions—no manipulation of the interior $\boldsymbol{\kappa}$ can change temperatures at points where the BC prescribes them.

02.2.2 Transformation Thermotics: Power and Limits

Transformation thermotics exploits the form-invariance of Eq. (02.1) under coordinate transformations. A mapping $\mathbf{x} \rightarrow \mathbf{x}'$ transforms the conductivity tensor as:

$$\boldsymbol{\kappa}' = \frac{\mathbf{J} \cdot \boldsymbol{\kappa} \cdot \mathbf{J}^T}{\det(\mathbf{J})} \quad (02.2)$$

At steady state, transformation thermotics can: redirect flux (cloaking), concentrate flux, rotate flux direction, invert flux locally, and engineer anisotropic patterns ($\kappa_{\parallel} \neq \kappa_{\perp}$).

Steady-State Limitation

The Fundamental Steady-State Limitation: In steady state with fixed boundary conditions, transformation thermotics controls the *direction and magnitude* of heat flux $\mathbf{q}$ but cannot change the absolute temperature at any point where the temperature is prescribed by BCs. The maximum principle for elliptic PDEs guarantees that temperature extrema occur on the boundary, not in the interior.

Implication: A thermal cloak makes an object thermally “invisible” by restoring the external field, but the cloaked object’s temperature is *not* controlled—it equilibrates to whatever the rerouted flux pattern dictates. Claims of steady-state “thermal shielding” (reducing cloaked-object temperature) are physically incorrect for passive metamaterial cloaks.

02.2.3 Implications for Cloaking Claims

Table 02.6: Steady-state capabilities and limitations of thermal cloaks.

Claim	Achievable?	Mechanism
Restore external T field	Yes	Flux rerouting
Eliminate thermal signature	Yes	Pattern restoration
Reduce object temperature	No	Maximum principle
Create thermal “cold spot”	No	Maximum principle
Shield object from heat	No*	Steady-state limit

*Transient shielding *is* possible; see §02.3.

Design Rule:

DR-02.1 — Steady-State Limitation At steady state, transformation thermotics controls heat flux *direction and magnitude* but not absolute temperature. If the design objective requires changing the temperature at a specific location, one must (a) change boundary conditions, (b) operate in the transient regime (§02.3), or (c) use active feedback control.

Trilogy Connection:

Eikonal Bridge Ch. 3 — Optical Cloaking Analogy An optical cloak based on transformation optics restores the *phase front* of the external field but cannot reduce optical intensity inside the cloaked region to zero. In both thermal and optical cloaking, the transformation controls the *field pattern* but not the *field amplitude* within the transformed region.

02.2.4 Worked Example: WE-02.1—Steady-State Cloak Temperature

Case Study:

WE-02.1: Temperature Inside a Steady-State Thermal Cloak **Problem:** A cylindrical thermal cloak ($R_1 = 5$ mm, $R_2 = 15$ mm) in a silicon substrate has boundary temperatures $T_{\text{hot}} = 400$ K (left) and $T_{\text{cold}} = 300$ K (right), separated by $W = 60$ mm. What is the temperature at the cloak center at steady state?

Solution: In virtual space, the unperturbed field is linear:

$$T_{\text{center}} = 400 - \frac{100}{60} \times 30 = 350 \text{ K} \quad (02.3)$$

The cloak reroutes flux but the center equilibrates to 350 K—identical to the no-cloak case. To reduce the temperature below 350 K requires changing the BCs or exploiting transient effects (§02.3).

Steady-State Limitation

Myth Audit: Common Steady-State Misconceptions

DR-02.1 has direct implications for evaluating claims in the thermal metamaterial literature. The following “red flags” help engineers audit published or proposed designs:

Red Flag 1: “The thermal cloak reduces the protected region’s temperature below ambient.”

Rebuttal: At steady state with no internal heat source, the maximum principle guarantees that interior temperatures lie between the boundary extremes. A passive cloak reroutes flux but cannot create a cold spot below the minimum boundary temperature. If such cooling is observed experimentally, it is a transient artifact ($t < \tau_{th}$) that will vanish at equilibrium.

Red Flag 2: “The metamaterial shield blocks thermal radiation from reaching the interior.”

Rebuttal: At steady state, a passive structure in thermal equilibrium emits as much as it absorbs (Kirchhoff’s law). “Blocking” implies a permanent temperature reduction, which requires either active cooling or boundary-condition modification—not a passive metamaterial. The shield can *redirect* flux, reducing it along certain paths while increasing it along others, but the total thermal budget is conserved.

Red Flag 3: “Our concentrator achieves 50% higher temperature at the focus than the boundary temperature.”

Rebuttal: A thermal concentrator increases *flux density* (W/m^2), not temperature. At steady state without internal sources, the maximum temperature cannot exceed the hottest boundary. Flux concentration is real and useful (e.g., for thermoelectric harvesting), but conflating flux density with temperature is a category error.

Red Flag 4: “Simulation shows the device works; we used COMSOL with 60-second solve time.”

Rebuttal: If the simulation used transient analysis, a 60-second simulated duration may not be sufficient to reach steady state (check $Fo > 3$). If it used steady-state analysis, the solver finds the correct equilibrium but may mask the *transient* performance that the real system will exhibit during warm-up. Always verify: does the *production* use case match the simulation temporal regime?

Red Flag 5: “The metamaterial provides thermal insulation equivalent to a vacuum gap.”

Rebuttal: Transformation thermotics manipulates the *direction* of heat flux via anisotropic κ , but anisotropy implies both a high- κ and a low- κ principal axis. The low-axis resistance is bounded by the constituent materials’ bulk conductivities. True “insulation” requires low κ in *all* directions—a scalar property that transformation thermotics cannot create from high- κ constituents.

02.3 Transient Regime: The Window of Opportunity

02.3.1 The Full Heat Equation

The time-dependent heat equation:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\boldsymbol{\kappa} \cdot \nabla T) + \dot{q} \quad (02.4)$$

involves *two* material properties: $\boldsymbol{\kappa}$ (which transformation thermotics manipulates) and ρc_p (which it does *not* independently transform).

An important conceptual clarification: transient metamaterial control is *time borrowing*, not the creation of a new equilibrium. During the transient window ($t \lesssim \tau_{\text{th}}$), engineered $\boldsymbol{\kappa}$ distributions can produce spatial temperature gradients—including local cooling below ambient levels along specific paths—that are impossible at steady state (DR-02.1). However, once $t \gg \tau_{\text{th}}$, the system inevitably returns to the boundary-determined steady-state temperature field. The metamaterial buys time; it does not rewrite the thermodynamic endpoint.

02.3.2 Thermal Diffusivity and Characteristic Timescales

$$\alpha_{\text{th}} = \frac{\kappa}{\rho c_p}, \quad \tau_{\text{th}} = \frac{L^2}{\alpha_{\text{th}}} \quad (02.5)$$

Table 02.7: Thermal diffusivity and characteristic time constant for semiconductor-relevant materials at 300 K ($L = 1$ mm for τ_{th} column).

Material	κ (W m ⁻¹ K ⁻¹)	α_{th} (mm ² /s)	τ_{th} ($L = 1$ mm)
Silicon	148	88	11 μ s
Copper	401	117	8.5 μ s
Germanium	60	34	29 μ s
SiO ₂ (fused)	1.4	0.83	1.2 ms
As ₂ Se ₃	0.24	0.12	8.3 ms
BT resin	0.3	0.18	5.6 ms
Air (1 atm)	0.026	22	45 μ s

Equivalence of Fo and $\tau_{\text{proc}}/\tau_{\text{th}}$. The Fourier number is defined as $\text{Fo} = \alpha_{\text{th}} t / L^2 = t / \tau_{\text{th}}$. When evaluated at the process timescale $t = \tau_{\text{proc}}$, it yields:

$$\text{Fo}|_{t=\tau_{\text{proc}}} = \frac{\tau_{\text{proc}}}{\tau_{\text{th}}} \quad (02.6)$$

Thus the regime boundaries can be stated equivalently as Fourier-number thresholds or $\tau_{\text{proc}}/\tau_{\text{th}}$ thresholds—they are the same quantity. This chapter uses $\tau_{\text{proc}}/\tau_{\text{th}}$ throughout because it directly compares two engineer-specified quantities (the process duration and the system's intrinsic thermal response time), making physical interpretation immediate. The Fo notation is retained for compatibility with the heat-transfer literature (particularly when t varies continuously rather than being fixed at τ_{proc}).

The Fourier number and Biot number:

$$\text{Fo} = \frac{\alpha_{\text{th}} t}{L^2} = \frac{t}{\tau_{\text{th}}}, \quad \text{Bi} = \frac{h L}{\kappa} \quad (02.7)$$

02.3.3 The Transient Advantage: Local Temperature Manipulation

Transient Regime

The Window of Opportunity: During the transient regime ($Fo \lesssim 1$), a thermal metamaterial can achieve what is impossible at steady state: (a) **delay** heat penetration (transient shielding), (b) **accelerate** local equilibration via high- α_{th} pathways, (c) **create** temporary temperature differences, and (d) **shape** the spatial profile during equilibration. All these capabilities vanish once steady state is reached.

The thermal diffusion length sets the spatial reach:

$$L_{diff} = \sqrt{\alpha_{th} \tau} \quad (02.8)$$

For a 1 mm As_2Se_3 layer: $t_{shield} \approx d^2/\alpha_{th} = 8.3$ ms. Compare silicon: $t_{shield} = 11 \mu s$ —a $700\times$ difference that maps directly onto the landscape of Table 02.4.

Design Rule:

DR-02.2 — Transient Control Criterion Passive metamaterial transient control requires $\tau_{proc} \lesssim \tau_{th}$, where $\tau_{th} = L^2/\alpha_{th}$. Design metamaterial thickness $L \geq \sqrt{\alpha_{th} \cdot \tau_{proc}}$.

02.3.4 Dynamic Γ : Mode Dominance in Transient

During RTP, Γ becomes time-dependent through:

$$G_{rad}(t) = 4 \varepsilon \sigma_{SB} T(t)^3 A \quad (02.9)$$

For an RTP ramp 300 K $\rightarrow$ 1273 K: $G_{rad}(1273)/G_{rad}(300) = (1273/300)^3 \approx 76$.

Multiphysics Integration

Dynamic Mode-Path Matrix: During rapid thermal processing, the Mode-Path Matrix must be evaluated at multiple temperature snapshots. A path classified as conduction-dominated at 300 K ($\Gamma \sim 10^{-5}$) may become mixed-mode at 800 K ($\Gamma \sim 10^{-2}$) and approach radiation dominance at 1273 K ($\Gamma \sim 10^{-1}$). Tool selection must use the *worst-case* Γ along the trajectory.

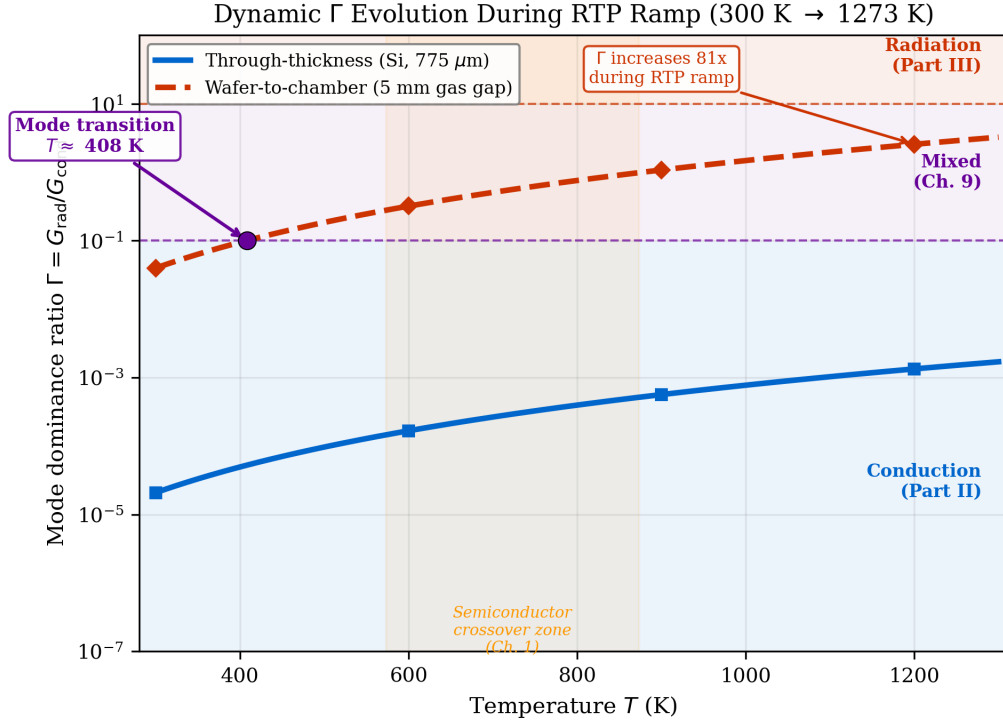


Figure 02.3: Dynamic Γ during RTP from 300 K to 1273 K. The through-thickness path (solid blue) remains conduction-dominated. The wafer-to-chamber path (dashed red) crosses into mixed mode above ~ 800 K. The Mode-Path Matrix must be evaluated at multiple snapshots.

Multiphysics Integration

Recipe Card: Computing Dynamic $\Gamma(t)$ for Transient Processes

When $\tau_{\text{proc}}/\tau_{\text{th}}$ falls in the intermediate regime (0.1–10), Γ evolves significantly during the process. Use this procedure to map the Γ trajectory and select appropriate design tools.

R-1: Define the thermal path. Identify the heat-flow path of interest (e.g., “through wafer thickness,” “wafer surface to chamber wall”). Record geometry: cross-section area A , path length L , gap distance d (if radiative).

R-2: Specify the temperature range. Determine $T_{\min}$ and $T_{\max}$ for the process. Select 4–6 evaluation temperatures T_j spanning this range, including the endpoints and any known transition points (e.g., Si κ knee near 500 K, phase-change temperatures).

R-3: Compute $G_{\text{cond}}(T_j)$ at each snapshot.

$$G_{\text{cond}}(T_j) = \frac{\kappa(T_j) A}{L}$$

Use temperature-dependent $\kappa(T)$ data. For silicon: $\kappa \approx 150 (300/T)^{1.3} \text{ W m}^{-1} \text{ K}^{-1}$ above 300 K.

R-4: Compute $G_{\text{rad}}(T_j)$ at each snapshot.

$$G_{\text{rad}}(T_j) = 4 \varepsilon \sigma_{\text{SB}} T_j^3 A$$

Use effective emissivity ε for the surface pair. For enclosed geometries, use the radiation-network emissivity (Chapter 1, §01.2).

R-5: Compute $\Gamma(T_j) = G_{\text{rad}}(T_j)/G_{\text{cond}}(T_j)$. Plot Γ vs. T on a logarithmic y -axis. Mark the regime boundaries $\Gamma = 0.1$ and $\Gamma = 10$.

R-6: Identify the peak Γ and any mode transitions.

- If $\Gamma < 0.1$ throughout: conduction-dominated at all times $\rightarrow$ use Part II tools (Chs. 5–7).
- If $\Gamma > 10$ throughout: radiation-dominated at all times $\rightarrow$ use Part III tools (Chs. 8–11).
- If Γ crosses 0.1 or 10 during the process: **mode transition occurs**. Design must account for both modes. Use the *worst-case* (highest) Γ for tool selection to ensure the radiation pathway is not neglected.

R-7: Determine snapshot count for MPM. As a minimum: evaluate the Mode-Path Matrix (Chapter 4) at $T_{\min}$, $T_{\max}$, and at every mode-transition crossing identified in Step R-6. For smooth Γ trajectories without crossings, three snapshots (start, midpoint, end) are sufficient. For trajectories with crossings, add one snapshot within ± 50 K of each crossing temperature.

Linearization guidance: Steps R-3 and R-4 use point evaluations at each T_j . This is valid when the temperature excursion per snapshot interval is $\Delta T \lesssim 100$ K. For larger intervals, subdivide further or use integral-averaged conductances (Chapter 4, §04.7).

Quick sanity check: G_{rad} scales as T^3 ; G_{cond} for Si scales as $T^{-1.3}$. Therefore $\Gamma \propto T^{4.3}$ —it increases by a factor of $\sim 80\times$ between 300 K and 1273 K for the same geometry. Any RTP-class process will exhibit significant Γ evolution.

02.4 Quasi-Steady State: The Practical Operating Point

02.4.1 Cyclic Processes in Semiconductor Fabrication

A system is in **quasi-steady state** when:

$$\frac{\Delta T_{\text{cycle}}}{T_{\text{mean}} - T_{\text{ambient}}} \ll 1 \quad (02.10)$$

Design Rule:

DR-02.3 — Quasi-Steady-State Classification For cyclic processes, compute $\tau_{\text{proc}}/\tau_{\text{th}}$. If > 10 : steady-state tools. If < 0.1 : time-averaged steady-state with $\bar{q}$. For 0.1–10: full transient analysis; C_{th} is a critical design parameter.

02.4.2 Time-Averaged vs. Instantaneous Profiles

The thermal penetration depth at cycling frequency f :

$$\delta_{\text{pen}} = \sqrt{\frac{\alpha_{\text{th}}}{\pi f}} \quad (02.11)$$

Table 02.8: Thermal penetration depth for selected materials and process frequencies at 300 K.

Material	α_{th} (mm ² /s)	$\delta_{\text{pen}}@1$ Hz	$\delta_{\text{pen}}@100$ Hz	$\delta_{\text{pen}}@10$ kHz
Silicon	88	5.3 mm	0.53 mm	53 μm
SiO ₂	0.83	0.51 mm	51 μm	5.1 μm
As ₂ Se ₃	0.12	0.20 mm	20 μm	2.0 μm
Germanium	34	3.3 mm	0.33 mm	33 μm
Copper	117	6.1 mm	0.61 mm	61 μm

02.4.3 Worked Example: WE-02.2—ALD Cycle Classification

Case Study:

WE-02.2: ALD Cycle Thermal Regime **Problem:** ALD deposits Al₂O₃ at 300 °C on a 300 mm Si wafer. Cycle time $\tau_{\text{proc}} = 4.2$ s. Chuck contact $h_c = 500$ W/m²K. Classify the regime.

Solution:

$$\tau_{\text{th}} = L^2/\alpha_{\text{th}} = (775 \times 10^{-6})^2/(88 \times 10^{-6}) = 6.8 \text{ ms.}$$

$$\tau_{\text{proc}}/\tau_{\text{th}} = 4.2/0.0068 = 620 \gg 10. \text{ Quasi-steady with full equilibration.}$$

$$\delta_{\text{pen}} = \sqrt{88/(0.24\pi)} = 10.8 \text{ mm} \gg \text{wafer thickness.}$$

$$\text{Bi} = 500 \times 7.75 \times 10^{-4}/148 = 2.6 \times 10^{-3} \ll 1. \text{ Lumped capacitance valid.}$$

Key: The thermal challenge is the wafer–chuck interface, not the wafer bulk.

02.4.4 Worked Example: WE-02.4—LWIR Ge Lens Power-On Transient

Case Study:

WE-02.4: LWIR Germanium Lens Module Warm-Up Problem: A germanium LWIR lens element ($D = 50$ mm, thickness $t = 8$ mm, $\alpha_{\text{th}} = 34$ mm²/s, $dn/dT = +4.0 \times 10^{-4}$ K⁻¹) is mounted in an aluminum housing. After power-on, the housing warms from $T_0 = 20^\circ\text{C}$ to $T_f = 35^\circ\text{C}$ over $\tau_{\text{proc}} \approx 20$ s. The lens absorbs 0.5 W from the detector/electronics heat path. Classify the temporal regime, estimate the focus drift, and determine when the lens reaches its “operational steady state.”

Solution:**Step 1: Thermal time constant.**

Radial thermal time constant (heat enters from housing rim):

$$\tau_{\text{th}} = \frac{R^2}{\alpha_{\text{th}}} = \frac{(25 \times 10^{-3})^2}{34 \times 10^{-6}} = 18.4 \text{ s} \quad (02.12)$$

Step 2: Timescale ratio. $\tau_{\text{proc}}/\tau_{\text{th}} = 20/18.4 = 1.1$. **Intermediate regime**—per DR-02.3 (Table 02.4, “LWIR Ge lens power-on”), full transient analysis is required.

Step 3: Focus drift during transient.

During warm-up, the lens develops a radial temperature gradient (warmer rim, cooler center) because heat enters from the housing. The resulting thermal focal power (Chapter 1, Eq. 01.24):

$$\Phi_{\text{thermal}} = \frac{dn/dT}{2\pi\kappa R^2} \frac{P_{\text{abs}}}{2\pi \times 60 \times (25 \times 10^{-3})^2} = \frac{4.0 \times 10^{-4} \times 0.5}{2\pi \times 60 \times (25 \times 10^{-3})^2} = 0.85 \text{ m}^{-1} \quad (02.13)$$

At $\lambda = 10$ μm , the thermal OPD during peak transient:

$$W_{\text{PV}} \approx (dn/dT) \cdot \Delta T \cdot t = 4.0 \times 10^{-4} \times 5 \times 8 \times 10^{-3} = 16 \text{ } \mu\text{m} \approx 1.5 \lambda \quad (02.14)$$

This exceeds the DR-01.3 threshold ($\lambda/10$) by a factor of 15—the lens is **badly out of spec** during the transient. Steady-state athermalization (which compensates only the uniform ΔT component) does not address this gradient-driven aberration.

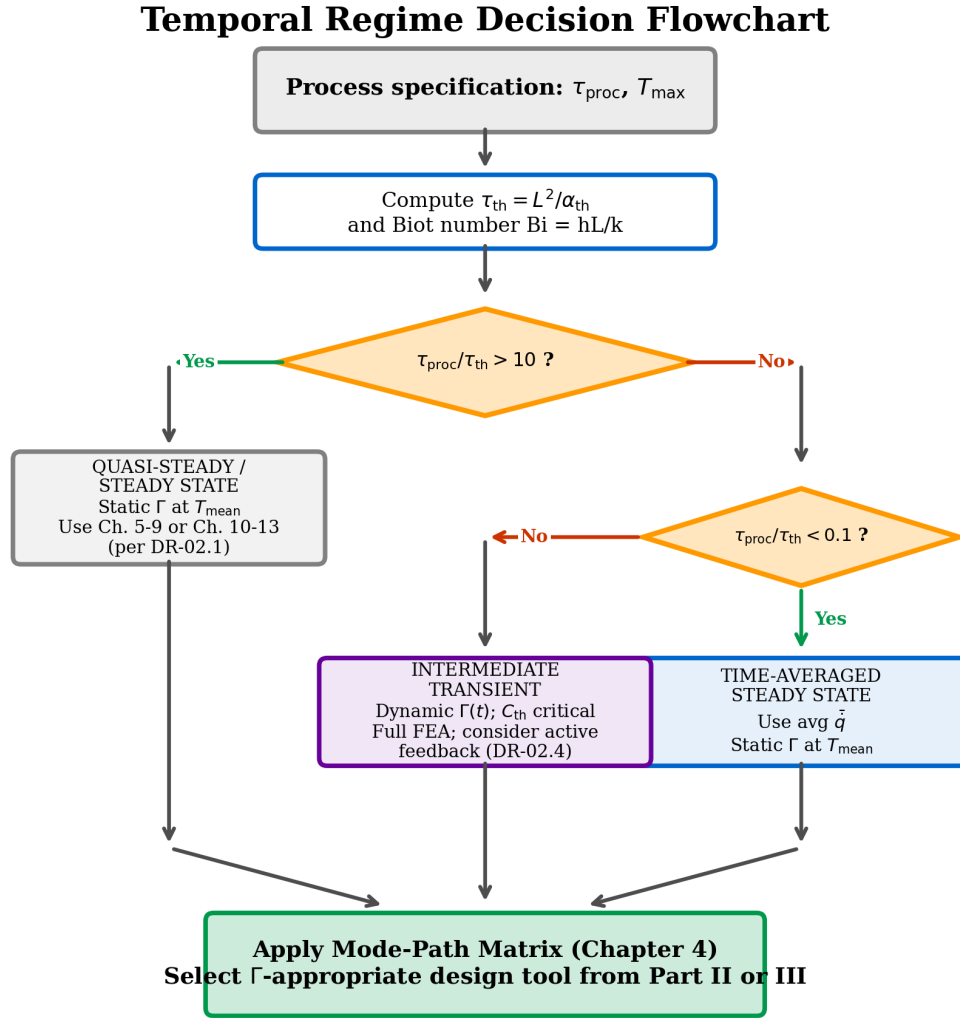
Step 4: Time to operational steady state.

The system approaches steady state for $t > 3\tau_{\text{th}} \approx 55$ s. During this period, thermal aberrations evolve from gradient-dominated (coma, astigmatism from asymmetric heating) to uniform-dominated (pure defocus), then are compensated by the athermalization design.

Key insight: LWIR systems routinely experience a 30–60 s “focus drift window” after power-on where steady-state athermalization is insufficient and active refocus or thermal pre-conditioning is needed. This is the Pattern 3 phenomenon identified in §02.1.1.

02.5 Matching Design to Temporal Regime

02.5.1 The Decision Framework



Design Rules: DR-02.3 (classification), DR-02.4 (active vs. passive), DR-T.8 (temporal compatibility)

Figure 02.4: Temporal regime decision flowchart. Compute τ_{th} , evaluate τ_{proc}/τ_{th} , and select the design approach. All branches feed into the Mode-Path Matrix (Chapter 4).

The flowchart logic of Figure 02.4 is restated here as a text-complete decision procedure so the reader can apply it without reference to the figure:

Design Rule: Text-Complete Decision Procedure

Inputs: Process timescale τ_{proc} , maximum process temperature T_{max} , characteristic length L , material thermal diffusivity α_{th} , and heat transfer coefficient h at the dominant boundary.

Step 1: Compute τ_{th} and Bi.

$$\tau_{\text{th}} = \frac{L^2}{\alpha_{\text{th}}}, \quad \text{Bi} = \frac{h L}{\kappa}$$

Use the L appropriate to the thermal path under analysis (see “Choosing L ” box in §02.1).

Step 2: Evaluate $\tau_{\text{proc}}/\tau_{\text{th}}$.

- If $\tau_{\text{proc}}/\tau_{\text{th}} > 10$: **Quasi-steady / Steady state.** Use static Γ evaluated at T_{mean} . Apply steady-state tools from Chapters 5–9 (conduction) or Chapters 10–13 (radiation) per DR-02.1.
- If $\tau_{\text{proc}}/\tau_{\text{th}} < 0.1$: **Time-averaged steady state.** Use time-averaged heat generation $\bar{q}$ and evaluate static Γ at T_{mean} . Thermal capacitance C_{th} is *not* a critical design parameter.
- If $0.1 \leq \tau_{\text{proc}}/\tau_{\text{th}} \leq 10$: **Intermediate / Full transient.** C_{th} is a critical design parameter. Evaluate dynamic $\Gamma(T)$ at multiple snapshots (see Recipe Card below). Consider active feedback per DR-02.4.

Step 3: Check Bi for internal gradient.

- If $\text{Bi} < 0.1$: lumped capacitance valid; internal gradients negligible.
- If $\text{Bi} > 0.1$: spatial temperature distribution must be resolved (FEA or analytical series).

Step 4: Feed into Mode-Path Matrix (Chapter 4). The regime classification and Γ value(s) determine which row of the Mode-Path Matrix applies to each thermal path. For the intermediate regime, the MPM must be evaluated at multiple time snapshots (DR-02.5, DR-T.8).

Table 02.9: Design tool selection by temporal regime.

Regime	Metamaterial Capability	Design Tool	Γ Treatment
Steady ($\text{Fo} \gg 1$)	Flux direction only	Ch. 5 transform.; static MPM	Static Γ at T_{op}
Transient ($\text{Fo} \lesssim 1$)	Local T control	Transient FEA; ρc_p critical	Dynamic $\Gamma(t)$
Quasi-steady	Time-averaged flux	Steady with $\bar{q}$	Static at T_{mean}
Intermediate	Partial transient	Full transient + active	Dynamic; evaluate at peak

02.5.2 Active vs. Passive Selection

Design Rule:

DR-02.4 — Active vs. Passive Selection Use passive metamaterial control when $\tau_{\text{proc}}/\tau_{\text{th}} > 10$ and the objective is flux management. Use active feedback for setpoint control or when $0.1 < \tau_{\text{proc}}/\tau_{\text{th}} < 10$. Hybrid designs (passive shaping + active setpoint) often provide the best performance-to-cost ratio.

02.5.3 Worked Example: WE-02.3—Laser Spike Annealing

Case Study:

WE-02.3: Laser Spike Annealing **Problem:** LSA heats the top 10 μm of a Si wafer from 400 K to 1573 K with dwell $\tau_{\text{proc}} = 0.5$ ms. Classify regime, evaluate transient shielding potential, and track dynamic Γ .

Solution:

Heated layer: $\tau_{\text{th}} = (10 \times 10^{-6})^2 / (88 \times 10^{-6}) = 1.1 \mu\text{s}$. $\tau_{\text{proc}}/\tau_{\text{th}} = 450 \gg 10$: heated layer equilibrates instantly.

Full wafer: $\tau_{\text{th}}^{\text{wafer}} = 6.8$ ms. $\tau_{\text{proc}}/\tau_{\text{th}} = 0.07 < 0.1$: **bulk wafer is transient.**

Diffusion length: $L_{\text{diff}} = \sqrt{88 \times 10^{-6} \times 0.5 \times 10^{-3}} = 210 \mu\text{m}$.

SiO₂ barrier (50 μm): $t_{\text{shield}} = (50 \times 10^{-6})^2 / (0.83 \times 10^{-6}) = 3.0$ ms $> \tau_{\text{proc}}$. Effective transient protection.

Dynamic Γ for wafer-to-chamber path: at 400 K: $\Gamma = 9.5 \times 10^{-3}$ (conduction). At 1573 K: $\Gamma = 0.58$ (**mixed mode**). Chapter 9 tools needed at peak temperature.

02.6 Temporal Considerations in Network Analysis

02.6.1 Dynamic Thermal Networks: Adding Capacitance

Each node carries thermal capacitance $C_{\text{th}} = \rho c_p V$. The node energy balance:

$$C_{\text{th}}^{(i)} \frac{dT_i}{dt} = \sum_j G_{ij}(T_j - T_i) + \dot{Q}_i \quad (02.15)$$

In matrix form:

$$\mathbf{C} \frac{d\mathbf{T}}{dt} + \mathbf{G} \mathbf{T} = \mathbf{Q}(t) \quad (02.16)$$

02.6.2 System Time Constants

Eigenvalues of $\mathbf{C}^{-1}\mathbf{G}$ yield:

$$\tau_k = 1/\lambda_k, \quad k = 1, \dots, N \quad (02.17)$$

The slowest mode τ_1 is the dominant system time constant.

Design Rule:

DR-02.5 — System Time Constant Compare τ_1 with τ_{proc} . If $\tau_1 > \tau_{\text{proc}}$, transient analysis is mandatory at the system level (equivalent to DR-T.8).

02.6.3 Critical Path Dynamics

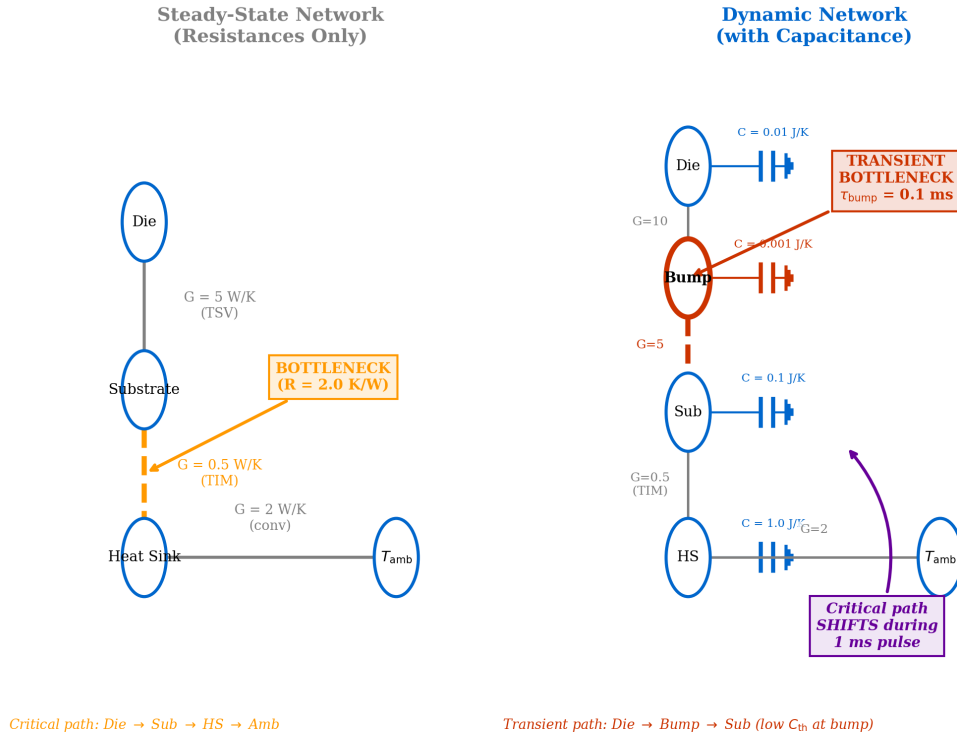


Figure 02.5: Thermal network with capacitance. **Left:** Steady-state (resistances only); bottleneck at TIM. **Right:** Dynamic network; during a 1 ms pulse, the solder bump (low C_{th}) becomes the transient bottleneck. The critical path *shifts*.

The spatial distribution of C_{th} along the critical path determines how the path shifts during transients: capacitance-dominated nodes buffer pulses while resistance-dominated nodes heat rapidly. The steady-state critical path may differ from the transient critical path.

02.6.4 Time-Dependent Conductances

Phase-change materials (VO_2 , $\text{Ge}_2\text{Sb}_2\text{Te}_5$) create temperature-dependent conductance steps. Silicon's $\kappa \propto T^{-1.3}$ above 300 K makes G_{cond} *decrease* during heating while G_{rad} *increases* ($\propto T^3$)—both effects push Γ higher during ramps.

Failure Mode:

Thermal Runaway via Positive Feedback Increasing T reduces G_{cond} (Si κ dropping) while heat source power remains constant—a positive feedback loop. The T^4 radiation increase eventually provides sufficient G_{rad} to stabilize. Identifying this crossover via the dynamic Γ trajectory is critical for RTP process safety.

Table 02.10: Complete set of temporal regime design rules and operational procedures (v3).

Rule	Section	Description	v3 Operational Support
DR-02.1	§02.2	Steady-state: flux direction control only. Temperature change requires BC modification, transient operation, or active feedback. (Maximum principle for $\dot{q} = 0$.)	<i>Myth Audit</i> (§02.2): five red-flag tests for auditing steady-state claims.
DR-02.2	§02.3	Transient control requires $\tau_{\text{proc}} \lesssim \tau_{\text{th}}$. Design barrier thickness $L \geq \sqrt{\alpha_{\text{th}} \cdot \tau_{\text{proc}}}$. Transient window is “time borrowing”—not a new equilibrium.	<i>Time-borrowing note</i> (§02.3): clarifies that transient gradients vanish for $t \gg \tau_{\text{th}}$.
DR-02.3	§02.4	Cyclic classification: $\tau_{\text{proc}}/\tau_{\text{th}} > 10$ (steady tools); < 0.1 (time-averaged steady); $0.1\text{--}10$ (full transient; C_{th} critical). Note: $\text{Fo} _{t=\tau_{\text{proc}}} = \tau_{\text{proc}}/\tau_{\text{th}}$ (Eq. 02.6).	<i>Choosing L</i> (§02.1): match L to transport path to avoid misclassification.
DR-02.4	§02.5	Passive metamaterial for $\tau_{\text{proc}}/\tau_{\text{th}} > 10$ + flux-direction goals. Active feedback for temperature setpoint control or intermediate regime.	<i>Decision Procedure</i> (§02.5): 4-step text-complete if/then logic with Bi check.
DR-02.5	§02.6	System $\tau_1 > \tau_{\text{proc}}$ requires transient analysis at the system level (equivalent to DR-T.8). Critical path may shift between steady-state and transient regimes.	<i>Problem 2.6</i> : RC-network exercise for eigenvalue extraction and critical-path shift.
<i>Operational Procedures (v3 additions)</i>			
Recipe	§02.3	Dynamic $\Gamma(t)$ Recipe Card. 7-step procedure: (R-1) define path, (R-2) specify T range, (R-3) compute $G_{\text{cond}}(T_j)$, (R-4) compute $G_{\text{rad}}(T_j)$, (R-5) compute $\Gamma(T_j)$, (R-6) identify peak Γ and mode transitions, (R-7) determine MPM snapshot count. Quick check: $\Gamma \propto T^{4.3}$ for Si $\Rightarrow \sim 80\times$ increase over 300–1273 K.	Supports DR-02.3 (intermediate regime) and DR-02.4 (tool selection). Feeds directly into Ch. 4 MPM construction.
Exec.	Roadmap	Executive Summary. Three quantities to remember: (1) Fo classifies temporal regime, (2) Bi determines surface-vs-interior uniformity, (3) $\tau_{\text{proc}}/\tau_{\text{th}}$ determines available metamaterial tools and whether Γ tracking is dynamic.	Entry point for the entire chapter; enables rapid triage before detailed analysis.

Looking Forward

This chapter established the temporal dimension of thermal engineering through two complementary lenses: the mathematical framework (Fo, Bi, $\tau_{\text{proc}}/\tau_{\text{th}}$) and the application landscape spanning fourteen orders of magnitude in timescale. The three patterns identified in §02.1.1—silicon’s quasi-steady dominance, quantum systems’ extreme temporal range, and classical IR optics’ intermediate-regime residence—provide the engineering context for every subsequent chapter.

Chapter 3 pushes beyond the continuum description, addressing where Fourier’s law and far-field radiation both break down: fluctuation electrodynamics and near-field heat transfer at sub-100 nm gaps. Chapter 4 then integrates dual modes (Chapter 1), temporal regimes (Chapter 2), and near-field effects (Chapter 3) into the Mode-Path Matrix.

Research Roadmap

Table 02.11: Research roadmap for Chapter 2 topics.

Topic	Current State	5-Year Horizon	TPD Connection
Transient thermal metamaterials	Lab cloak demos with engineered ρc_p	Integrated transient shields in packaging	Ch. 5, Ch. 15
Dynamic Γ tracking	Manual per-snapshot calc	Automated $\Gamma(t)$ in FEA	Ch. 4, Ch. 15
Phase-change switches	VO ₂ demos; limited cycling	$> 10^6$ cycle switches	Ch. 6, Ch. 8
Thermal capacitance engineering	Ad hoc thermal mass	Designed C_{th} for transient path control	Ch. 4, Ch. 14
LWIR athermalization under transient	Static compensation only	Dynamic refocus with thermal Zernike tracking	Ch. 1, Ch. 10–11

Problems

1. **Problem 2.1: Steady-State Limitation.** A cylindrical thermal concentrator ($R_1 = 3$ mm, $R_2 = 10$ mm) has $T_{\text{hot}} = 500$ K and $T_{\text{cold}} = 300$ K. (a) Maximum interior temperature at steady state? (b) Can the concentrator raise local flux density above the boundary flux? (c) Sketch the flux lines.
2. **Problem 2.2: Transient Shielding Design.** Protect a MEMS device from a 5 ms thermal pulse through Si. Specify: (a) barrier material from Table 02.7, (b) minimum thickness, (c) steady-state R_{th} penalty. Discuss the trade-off.
3. **Problem 2.3: ALD vs. CVD Regime.** Compare (a) ALD at 1 Hz and (b) PE-CVD at 500 Hz RF cycling on a 300 mm Si wafer. For each: τ_{th} , $\tau_{\text{proc}}/\tau_{\text{th}}$, regime, δ_{pen} , design approach.
4. **Problem 2.4: Dynamic Γ Trajectory.** RTP ramp 300 K $\rightarrow$ 1273 K on Si. (a) Γ at 300, 500, 800, 1000, 1273 K for through-thickness path. (b) Repeat for wafer-to-chamber path (5 mm gas gap). (c) Crossover temperature. (d) Sketch $\Gamma(T)$ on log scale and locate on the landscape (Fig. 02.2).
5. **Problem 2.5: EUV Scanner Pulses.** 50 kHz EUV pulses deposit 1 mJ per pulse into 100 nm resist on Si. (a) Per-pulse ΔT in resist. (b) τ_{th} for resist layer. (c) τ_{th} for full wafer. (d) Quasi-steady over full field exposure?
6. **Problem 2.6: Thermal Network Eigenvalues.** 3-node flip-chip: Die ($C_{\text{th}} = 0.01$ J/K), Substrate (0.1 J/K), Heat sink (1 J/K). $G_{12} = 5$, $G_{23} = 0.5$, $G_{3\infty} = 2$ W/K. (a) $\mathbf{C}$, $\mathbf{G}$ matrices. (b) Eigenvalues and τ_k . (c) Steady state for $\tau_{\text{proc}} = 0.1$ s? (d) Transient bottleneck node.
7. **Problem 2.7: LWIR Lens Focus Drift.** A Ge LWIR lens ($D = 50$ mm, $t = 5$ mm, $\alpha_{\text{th}} = 34$ mm²/s, $dn/dT = +4.0 \times 10^{-4}$ K⁻¹) absorbs 0.3 W after power-on. (a) Compute τ_{th} . (b) Estimate peak thermal OPD during the transient. (c) Compare with DR-01.3 threshold. (d) How long until steady-state athermalization is valid? (e) Locate this scenario on the timescale landscape (Fig. 02.2) and verify it falls in the intermediate regime.
8. **Problem 2.8: Thermal Network Construction and Regime Classification.** A simplified flip-chip package consists of four thermal nodes connected in series: (1) silicon die ($\kappa = 150$ W m⁻¹ K⁻¹, $\rho c_p = 1.66 \times 10^6$ J m⁻³ K⁻¹, thickness 100 μ m, area 10 $\times$ 10 mm²), (2) thermal interface material (TIM: $\kappa = 5$ W m⁻¹ K⁻¹, $\rho c_p = 1.5 \times 10^6$ J m⁻³ K⁻¹, thickness 25 μ m), (3) copper substrate ($\kappa = 390$ W m⁻¹ K⁻¹, $\rho c_p = 3.4 \times 10^6$ J m⁻³ K⁻¹, thickness 1 mm), and (4) heat sink held at $T_{\text{amb}} = 300$ K via forced convection ($h = 500$ W m⁻² K⁻¹).
 - (a) Compute R_{th} and C_{th} for each of the four elements. Construct the 4×4 $\mathbf{G}$ and $\mathbf{C}$ matrices for the thermal network.
 - (b) Extract the eigenvalues λ_k of $\mathbf{C}^{-1}\mathbf{G}$ and compute the system time constants $\tau_k = 1/\lambda_k$. Identify the dominant (slowest) time constant τ_1 .
 - (c) A 10 ms laser annealing pulse deposits 50 W into the die. Compute Fo and Bi for the die, and classify the process using DR-02.3.
 - (d) Compare the steady-state critical path (highest R_{th} element) with the transient bottleneck (lowest C_{th} element). Does the critical path shift between the two cases? Explain why, referencing DR-02.5 and Figure 02.5.

- (e) For a continuous (steady-state) power dissipation of 5 W, re-evaluate $\tau_{\text{proc}}/\tau_{\text{th}}$ and confirm the regime classification. How does this change the design tool selection from Table 02.9?

Hint: The TIM dominates R_{th} at steady state, but the die (lowest C_{th}) heats fastest during the pulse—illustrating the critical-path shift described in §??.

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